

Fabrizio Malatesta

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EDUCATION:

University of California, San Diego (UCSD) - B.S in Biology w/ Specialization in Bioinformatics

- Minor in Computer Science | GPA/Major GPA: 3.8/4.0 | Provost Honors | Graduation: June 2027

RESEARCH EXPERIENCE:

Sanchez-Roige Lab - UCSD, La Jolla, CA | *Undergraduate Researcher* **Fall 2025-Present**

- Leading independent project on the genetic architecture of wisdom: GWAS and population-level genomic analyses (~131K participants), PGS analyses (~200K All of Us participants).
- **Presentations:** WCBSURC 2026 @ Sonoma State (Oral, Honorable Mention); URC 2026 @ UCSD; MURALS 2026 @ UCSD.

McVicker Lab - Salk Institute for Biological Studies, La Jolla, CA | *Undergraduate Researcher* **Summer 2025**

- Developed a Bash/Python pipeline to detect low-frequency off-target Cas9 edits from Superb-seq data, using guide homology scoring and statistical validation to filter technical artifacts.
- **Presentations:** NCRC 2026 @ Harvard; NCUR 2026 @ Richmond; UC LEADS Symposium @ UCSF; SRC 2025 @ UCSD.

Mueller Lab - Salk Institute for Biological Studies, La Jolla, CA | *Lab Tech I* **Fall 2024-June 2025**

- Conducted experiments on arbuscular mycorrhiza symbiosis, utilizing molecular techniques such as PCR, Gel electrophoresis, spore inoculation, microscopy analysis, and gene extraction.
- Designed an independent peptide treatment study across multiple plant species; presented to lab.

LEADERSHIP & ACTIVITIES:

UC LEADS Scholar – University of California, San Diego | **2025-2027**

- Selected for a competitive UC-wide program preparing scholars for STEM doctoral study, which includes a guaranteed Ph.D. fellowship; panelist advising prospective transfer students on research.

TRELS Scholar (Triton Research and Experiential Learning Scholars) - UCSD | **Winter 2026**

- Awarded \$1,000 in competitive university funding to support independent undergraduate research.

The Catalyst, UCSD's Undergraduate Research Journal | *Board Member - Outreach* **Spring 2026-Present**

- Oversee outreach initiatives connecting undergraduate researchers across UCSD with publication opportunities and demystifying the academic publishing process.

Saltman Quarterly | *Board Member - Outreach Committee* **May 2026-Present**

- Promote STEM to local K-12 students through interactive classroom experiments and field trips.

Undergraduate Bioinformatics Club (UBIC) | *Board Member - Finance* **Fall 2025-Present**

- Manage event finances and logistics, enabling bioinf. student body engagement and programming.

UCSD Mentor-Mentee Program | *Mentor* **Fall 2025-Present**

- Mentor newer undergraduate students navigating coursework, research, and university environment.

SKILLS:

- **Technical:** Python, R, Bash, Java, C++, Jupyter Notebooks, HPC (TSCC), LaTeX, FIJI/ImageJ.
- **Research:** GWAS & PGS analysis, single-cell sequencing analysis, LDSC, GenomicSEM, PCR, gel electrophoresis, gene extraction, experimental design, data interpretation.
- **Communication:** Scientific presentations, public speaking, mentoring, collaborative teamwork.
- **Languages:** Fluent in English and Italian; proficient in Spanish